Phase separation in multi-scale media - Two regimes

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Motivation

Composite materials are important to modern technology as the mixing of two different material properties at fine scales can give rise to unexpected emergent behavior. Therefore, understanding the process of phase separation on such materials is crucial to leveraging these processes for technological applications.

Main problem

- $\Omega \subset \mathbb{R}^N$ bounded open set with Lipschitz boundary.
- Q_1, Q_2 periodicity cells, i.e. domains having the paving property.
- $u \in W^{1,2}(\Omega;\mathbb{R}^M)$ distribution of phases.
- $W\colon \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}^M \to [0,+\infty)$ potential with fixed wells in $a, b \in \mathbb{R}^M$, Carathéodory, Q_1-periodic in first variable, Q_2-periodic in second variable, and linear growth at infinity.
- $\{\varepsilon_n\}_{n\in\mathbb{N}}$, $\{\delta_n\}_{n\in\mathbb{N}}$, $\{\eta_n\}_{n\in\mathbb{N}}$ infinitesimal sequences.

$$F_n(u) := \int_\Omega \left[W\left(\frac{x}{\delta_n}, \frac{x}{\eta_n}, u\right) + \varepsilon_n^2 |\nabla u|^2 \right] \mathrm{d}x \qquad u \in W^{1,2}(\Omega;\mathbb{R}^M),$$

possibly subject to a mass constraint:

$$\int_\Omega u \,\mathrm{d}x = m \in \mathbb{R}^M.$$

How to characterize the limits of minimizers of F_n as $n \to \infty$?

$\implies$ asymptotic Γ-expansion. There are five different regimes:

$$\eta_n \ll \delta_n \ll \varepsilon_n \qquad \eta_n \ll \varepsilon_n \ll \delta_n \qquad \varepsilon_n \ll \eta_n \ll \delta_n$$
$$\varepsilon_n \approx \eta_n \ll \delta_n \qquad \eta_n \ll \delta_n \approx \varepsilon_n$$

Zeroth-order and first-order Γ-expansion

$$F_n \overset{w-\Gamma(L^2)}{\longrightarrow} F_\infty^{(0)}(u) := \int_\Omega W_r(u)\,\mathrm{d}x \qquad u \in L^1(\Omega;\mathbb{R}^M).$$

The shape of W_r depends on the regime. In any case, the following hold:

- W_r potential with fixed wells in $a, b \in \mathbb{R}^M$;
- $\min F_\infty^{(0)} = 0$;
- energy of order ε_n.

Minimizers of $F_\infty^{(0)}$ are $\{u \in L^1(\Omega;\mathbb{R}^M) : u(x) \in \{a,b\} \text{ for a.e. } x \in \Omega\}$.
Not enough information on the limits of minimizers of F_n:
$\implies$ first-order Γ-development:

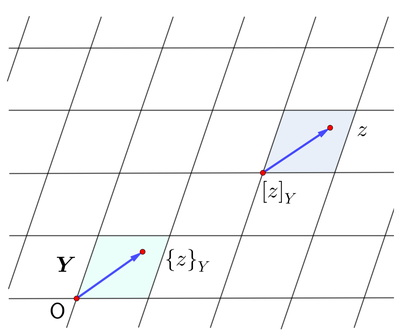
$$F_n^{(1)}(u) := \frac{F_n - \min F_\infty^{(0)}}{\varepsilon_n}$$
$$= \int_\Omega \left[\frac{1}{\varepsilon_n} W\left(\frac{x}{\delta_n}, \frac{x}{\eta_n}, u\right) + \varepsilon_n |\nabla u|^2 \right] \mathrm{d}x \qquad u \in W^{1,2}(\Omega;\mathbb{R}^M)$$

By defining $A := \{x \in \Omega : u(x) = a\}$, we get:

$$F_n^{(1)} \overset{\Gamma(L^1)}{\longrightarrow} F_\infty^{(1)}(u) := \int_{\partial^* A} \sigma(\nu_A(x)) \mathrm{d}\mathcal{H}^{N-1}(x) \qquad u \in \mathrm{BV}(\Omega;\{a,b\}),$$

where ν_A is the measure-theoretic exterior normal to the reduced boundary $\partial^* A$.
The formula for $\sigma(\nu)$ will change with the regime.

Two-scale periodic unfolding



Start from the identity:

$$z = \delta \left[\frac{z}{\delta}\right]_Y + \delta \left\{\frac{z}{\delta}\right\}_Y.$$

Instead of $\left\{\frac{z}{\delta}\right\}_Y$ have a variable $y \in Y$. In the same spirit define $\mathcal{G}_1\colon \Omega \times Q_1 \to \Omega$ and $\mathcal{G}_2\colon \Omega \times Q_1 \times Q_2 \to \Omega$ as

$$\mathcal{G}_1(x, y_1) := \delta \left[\frac{x}{\delta}\right]_{Q_1} + \delta y_1,$$
$$\mathcal{G}_2(x, y_1, y_2) := \delta \left[\frac{x}{\delta}\right]_{Q_1} + \eta \left[\frac{\delta y_1}{\eta}\right]_{Q_2} - \eta \iota_{2,\eta} + \eta y_2,$$

where $\iota_{2,\eta}$ accounts for the mismatch between micro-scales.

Define the first unfolding operator $\mathcal{U}_1\colon L^2(\Omega;\mathbb{R}^M) \to L^2(\Omega; L^2(Q_1;\mathbb{R}^M))$ as

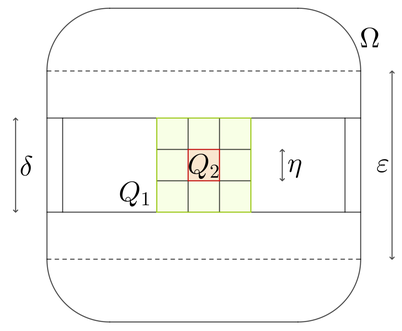
$$\mathcal{U}_1\varphi(x, y_1) := \varphi \circ \mathcal{G}_1(x, y_1) = \varphi\left(\delta \left[\frac{x}{\delta}\right]_{Q_1} + \delta y_1\right).$$

Define the second unfolding operator $\mathcal{U}_2\colon L^2(\Omega;\mathbb{R}^M) \to L^2(\Omega; L^2(Q_1; L^2(Q_2;\mathbb{R}^M)))$ as

$$\mathcal{U}_2\varphi(x, y_1, y_2) := \varphi \circ \mathcal{G}_2(x, y_1, y_2) = \varphi\left(\delta \left[\frac{x}{\delta}\right]_{Q_1} + \eta \left[\frac{\delta y_1}{\eta}\right]_{Q_2} - \eta \iota_{2,\eta} + \eta y_2\right).$$

Regime $\eta_n \ll \delta_n \ll \varepsilon_n$

$$\sigma(\nu) \equiv \sigma^{\mathrm{h}} := \inf_{\gamma \in \mathcal{A}(a,b)} \left\{ \int_{-1}^{1} 2\sqrt{W^{\mathrm{h}}(\gamma(t))}\,|\gamma'(t)|\,\mathrm{d}t \right\},$$



where W^{h} and $\mathcal{A}(a,b)$ are given by:

$$W^{\mathrm{h}}(z) := \int_{Q_1}\int_{Q_2} W(y_1, y_2, z)\,\mathrm{d}y_2\mathrm{d}y_1,$$
$$\mathcal{A}(a,b) := \{\gamma \in \mathrm{Lip}([-1,1];\mathbb{R}^M) : \gamma(-1) = a,\ \gamma(1) = b\}.$$

This is the only isotropic regime, since $\sigma(\nu)$ reduces to a constant:

$$F_\infty^{(1)}(u) = \sigma^{\mathrm{h}} \operatorname{Per}(\{u = a\};\Omega) \qquad u \in \mathrm{BV}(\Omega;\{a,b\}),$$

equivalent to the classic result by Modica-Mortola for the homogenized potential.
The idea is to first send $\delta_n \to 0$ to homogenize, and then send $\varepsilon_n \to 0$ to obtain the sharp interface for the homogenized potential.

- Liminf strategy: first lower the energy with unfolding operator

$$\liminf_{n\to\infty} F_n^{(1)}(u_n) \geq \liminf_{n\to\infty} \int_\Omega \left[\frac{1}{\varepsilon_n} \int_{Q_1}\int_{Q_2} W(y_1, y_2, \mathcal{U}_2 u_n)\mathrm{d}y_2\mathrm{d}y_1 + \varepsilon_n |\nabla u_n|^2 \right] \mathrm{d}x,$$

  then define an auxiliary potential $W^\xi(z)$ dependent on $\xi > 0$ such that $W^\xi(z) \to W^{\mathrm{h}}(z)$ uniformly as $\xi \to 0$, and further lower the energy

$$\liminf_{n\to\infty} F_n^{(1)}(u_n) \geq \liminf_{n\to\infty} \int_\Omega \sqrt{W^\xi(u_n(x))}|\nabla u_n(x)|\mathrm{d}x \geq \sigma^\xi \operatorname{Per}(\{u = a\};\Omega),$$

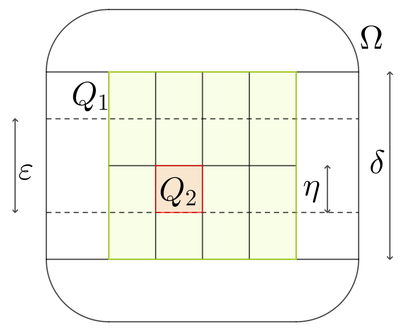
  and finally prove $\sigma^\xi \to \sigma^{\mathrm{h}}$ when $\xi \to 0$.
- Limsup strategy: construct a recovery sequence $\{u_n\}_{n\in\mathbb{N}}$ relative to the functional with $W^{\mathrm{h}}(z)$, and use an appropriate disjoint decomposition of the transition layer A_n and properties of the unfolding operator to prove that

$$\lim_{n\to\infty} \frac{1}{\varepsilon_n} \int_{A_n} \left| W\left(\frac{x}{\delta_n}, \frac{x}{\eta_n}, u_n(x)\right) - W^{\mathrm{h}}(u_n(x)) \right| \mathrm{d}x = 0,$$

  meaning that the micro-scales give no contribution to the energy in the limit.

Regime $\eta_n \ll \varepsilon_n \ll \delta_n$

$$\sigma(\nu) := \lim_{T\to\infty} \frac{1}{T^{N-1}} \int_{TQ_\nu \cap \partial E_\nu} \mathrm{d}_\eta^{\mathrm{h}}(y_1; a, b)\,\mathrm{d}\mathcal{H}^{N-1}(y_1),$$



where $\mathrm{d}_\eta^{\mathrm{h}}$ is given by:

$$\mathrm{d}_\eta^{\mathrm{h}}(y_1; p, q) := \inf_{\gamma \in \mathcal{A}(p,q)} \left\{ \int_{-1}^{1} 2\sqrt{W_\eta^{\mathrm{h}}(y_1, \gamma(t))}|\gamma'(t)|\mathrm{d}t \right\},$$
$$W_\eta^{\mathrm{h}}(y_1, z) := \int_{Q_2} W(y_1, y_2, z)\mathrm{d}y_2,$$

and $\mathcal{A}(p,q)$ is defined as before.
Here Q_ν is a unit cube with two faces orthogonal to ν, and E_ν is a plane-like minimizer relative to ν.

The problem amounts to a growing cell problem on the hyperplane tangent to the sharp interface. The idea is to first send $\varepsilon_n \to 0$ to obtain the sharp interface, and then send $\delta_n \to 0$ to homogenize on the interface.

- Liminf strategy: first bypass the dependence on y_2 by defining an auxiliary potential $W_\eta^\xi(y_1, z)$ such that $W_\eta^\xi \to W_\eta^{\mathrm{h}}$ uniformly as $\xi \to 0$, to get

$$\liminf_{n\to\infty} F_n^{(1)}(u_n) \geq \liminf_{n\to\infty} \int_\Omega 2\sqrt{W_\eta^\xi\left(\frac{x}{\delta_n}, u_n\right)}\,|\nabla u_n|\mathrm{d}x,$$

  and then use blow-up and techniques similar to first regime to deduce

$$\lim_{m\to\infty} \int_{Q_\nu} \sqrt{W_\eta^\xi(L_m y_1, v_m(y_1))}\,|\nabla v_m(y_1)|\mathrm{d}y_1 \geq \liminf_{m\to\infty} \int_{Q_\nu \cap \partial E_\nu} \mathrm{d}_m^\xi(L_m y_1; a, b)\,\mathrm{d}\mathcal{H}^{N-1}(y_1),$$

  and finally prove that this last integral converges to $\sigma(\nu)$ as $\xi \to 0$.
- Limsup strategy: approximate the interface and plane-like minimizers with polyhedral sets, and define the recovery sequence with a gluing construction. As before, the contribution of the smaller scale is proven to be negligible.

Ongoing research

- Remaining regimes: $\varepsilon_n \ll \eta_n \ll \delta_n$, $\varepsilon_n \approx \eta_n \ll \delta_n$, $\eta_n \ll \delta_n \approx \varepsilon_n$. [Cristoferi]
- Sharp interface dynamics for periodic domains:

$$-\varepsilon_n \triangle u + \frac{1}{\varepsilon_n} \partial_z W\left(\frac{x}{\delta_n}, u\right) = \lambda_n,$$

  with λ_n Lagrange multiplier corresponding to the mass constraint. Study of the convergence of the gradient flows of the sharp interface dynamic as $n \to \infty$. [Cristoferi, Ganedi]